



MATHTEACH: Fractions and Proportional Reasoning

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Learning Objectives

- Add, subtract, multiply, and divide fractions and mixed numbers
- Connect fractions, decimals, and percents
- Set up and solve proportions
- Interpret ratios and unit rates in real-world contexts

Simplify each expression completely. Show all steps and circle your final answer.

Adding fractions

1. Add the fractions and simplify: $\frac{1}{2} + \frac{3}{3}$.

$$\frac{1}{2} + \frac{3}{3}$$

Answer: _____

2. A recipe calls for $\frac{1}{4}$ cup of flour and $\frac{3}{3}$ cup of sugar. How much total dry ingredient is needed?

$$\frac{1}{4} + \frac{3}{3}$$

Answer: _____

3. Add: $\frac{4}{7} + \frac{5}{8}$. Simplify your answer completely.

$$\frac{4}{7} + \frac{5}{8}$$

Answer: _____

4. Add the fractions and simplify: $\frac{5}{2} + \frac{4}{7}$.

$$\frac{5}{2} + \frac{4}{7}$$

Answer: _____

5. A recipe calls for $\frac{1}{4}$ cup of flour and $\frac{3}{6}$ cup of sugar. How much total dry ingredient is needed?

$$\frac{1}{4} + \frac{3}{6}$$

Answer: _____



6. Add: $\frac{5}{4} + \frac{2}{8}$. Simplify your answer completely.

$$\frac{5}{4} + \frac{2}{8}$$

Answer: _____

7. Add the fractions and simplify: $\frac{1}{4} + \frac{4}{5}$.

$$\frac{1}{4} + \frac{4}{5}$$

Answer: _____

8. A recipe calls for $\frac{2}{6}$ cup of flour and $\frac{3}{2}$ cup of sugar. How much total dry ingredient is needed?

$$\frac{2}{6} + \frac{3}{2}$$

Answer: _____

Multiplying fractions

9. Multiply the fractions and simplify: $\frac{3}{5} \times \frac{4}{4}$.

$$\frac{3}{5} \times \frac{4}{4}$$

Answer: _____

10. Multiply: $\frac{1}{5} \times \frac{1}{7}$. Simplify your answer.

$$\frac{1}{5} \times \frac{1}{7}$$

Answer: _____

11. A recipe calls for $\frac{2}{2}$ cup of sugar. If you are making $\frac{2}{4}$ of the recipe, how many cups of sugar do you need?

$$\frac{2}{2} \times \frac{2}{4}$$

Answer: _____

12. Multiply the fractions and simplify: $\frac{2}{2} \times \frac{6}{5}$.

$$\frac{2}{2} \times \frac{6}{5}$$

Answer: _____



13. Multiply: $\frac{3}{4} \times \frac{5}{4}$. Simplify your answer.

$$\frac{3}{4} \times \frac{5}{4}$$

Answer: _____

14. A recipe calls for $\frac{2}{2}$ cup of sugar. If you are making $\frac{3}{2}$ of the recipe, how many cups of sugar do you need?

$$\frac{2}{2} \times \frac{3}{2}$$

Answer: _____

Percent of a number

15. Find 75% of 120.

75% of 120

Answer: _____

16. A shirt costs \$120. It is 10% off. How many dollars is the discount?

10% of 120

Answer: _____

17. A store is offering a 41% discount on an item priced at \$88. How many dollars is the discount?

41% of 88

Answer: _____

18. Find 75% of 60.

75% of 60

Answer: _____

19. A shirt costs \$200. It is 50% off. How many dollars is the discount?

50% of 200

Answer: _____

20. A store is offering a 40% discount on an item priced at \$51. How many dollars is the discount?

40% of 51

Answer: _____

Solving proportions



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21. Solve the proportion: $\frac{2}{4} = \frac{2}{x}$. Find x .

$$\frac{2}{4} = \frac{2}{x}$$

Answer: _____

22. A car travels 11 miles on 2 gallons of gas. How many miles can it travel on 6 gallons? Set up a proportion $\frac{2}{11} = \frac{6}{x}$ and solve.

$$\frac{2}{11} = \frac{6}{x}$$

Answer: _____

23. If 6 shirts cost \$37, how much do 7 shirts cost? Set up and solve a proportion.

$$\frac{6}{37} = \frac{7}{x}$$

Answer: _____

24. Solve the proportion: $\frac{4}{6} = \frac{3}{x}$. Find x .

$$\frac{4}{6} = \frac{3}{x}$$

Answer: _____

25. A car travels 19 miles on 1 gallons of gas. How many miles can it travel on 5 gallons? Set up a proportion $\frac{1}{19} = \frac{5}{x}$ and solve.

$$\frac{1}{19} = \frac{5}{x}$$

Answer: _____

26. If 2 shirts cost \$13, how much do 10 shirts cost? Set up and solve a proportion.

$$\frac{2}{13} = \frac{10}{x}$$

Answer: _____

Writing and simplifying ratios

27. Write the ratio 12 to 36 in simplest form.

12 : 36

Answer: _____



28. A bag contains 2 red marbles and 6 blue marbles. Write the ratio of red to blue in simplest form.

2 : 6

Answer: _____

29. Write the ratio 24 to 9 in simplest form.

24 : 9

Answer: _____

30. A bag contains 12 red marbles and 4 blue marbles. Write the ratio of red to blue in simplest form.

12 : 4

Answer: _____





Topics: Solving proportions, Writing and simplifying ratios, Percent of a number, Adding fractions, Multiplying fractions. All answers verified by independent computation.

Solutions



Adding fractions

1. Add the fractions and simplify: $\frac{1}{2} + \frac{3}{3}$.

$$\frac{1}{2} + \frac{3}{3}$$

→ Find a common denominator, then add numerators. Simplify if possible.

→ Result: $\frac{3}{2}$.

Answer: $= \frac{3}{2}$

2. A recipe calls for $\frac{1}{4}$ cup of flour and $\frac{3}{3}$ cup of sugar. How much total dry ingredient is needed?

$$\frac{1}{4} + \frac{3}{3}$$

→ Total = $\frac{1}{4} + \frac{3}{3} = \frac{5}{4}$ cups.

Answer: $= \frac{5}{4}$

3. Add: $\frac{4}{7} + \frac{5}{8}$. Simplify your answer completely.

$$\frac{4}{7} + \frac{5}{8}$$

→ Find the LCD of 7 and 8: LCD = 56.

→ Rewrite each fraction with denominator 56.

→ Add numerators and simplify: $\frac{67}{56}$.

Answer: $= \frac{67}{56}$

4. Add the fractions and simplify: $\frac{5}{2} + \frac{4}{7}$.

$$\frac{5}{2} + \frac{4}{7}$$

→ Find a common denominator, then add numerators. Simplify if possible.

→ Result: $\frac{43}{14}$.

Answer: $= \frac{43}{14}$

5. A recipe calls for $\frac{1}{4}$ cup of flour and $\frac{3}{6}$ cup of sugar. How much total dry ingredient is needed?

$$\frac{1}{4} + \frac{3}{6}$$

→ Total = $\frac{1}{4} + \frac{3}{6} = \frac{3}{4}$ cups.

Answer: $= \frac{3}{4}$

6. Add: $\frac{5}{4} + \frac{2}{8}$. Simplify your answer completely.

$$\frac{5}{4} + \frac{2}{8}$$

→ Find the LCD of 4 and 8: LCD = 8.

→ Rewrite each fraction with denominator 8.

→ Add numerators and simplify: $\frac{3}{2}$.

Answer: $= \frac{3}{2}$



7. Add the fractions and simplify: $\frac{1}{4} + \frac{4}{5}$.

$$\frac{1}{4} + \frac{4}{5}$$

→ Find a common denominator, then add numerators. Simplify if possible.

→ Result: $\frac{21}{20}$.

Answer: $\frac{21}{20}$

8. A recipe calls for $\frac{2}{6}$ cup of flour and $\frac{3}{2}$ cup of sugar. How much total dry ingredient is needed?

$$\frac{2}{6} + \frac{3}{2}$$

→ Total = $\frac{2}{6} + \frac{3}{2} = \frac{11}{6}$ cups.

Answer: $\frac{11}{6}$



Multiplying fractions

9. Multiply the fractions and simplify: $\frac{3}{5} \times \frac{4}{4}$.

$$\frac{3}{5} \times \frac{4}{4}$$

→ Multiply numerators: $3 \times 4 = 12$.

→ Multiply denominators: $5 \times 4 = 20$.

→ Simplify $12/20$: $3/5$.

Answer: $\frac{12}{20} = 3/5$

10. Multiply: $\frac{1}{5} \times \frac{1}{7}$. Simplify your answer.

$$\frac{1}{5} \times \frac{1}{7}$$

→ Multiply numerators: $1 \times 1 = 1$.

→ Multiply denominators: $5 \times 7 = 35$.

→ Simplify $1/35$: $1/35$.

Answer: $\frac{1}{35} = 1/35$

11. A recipe calls for $\frac{2}{2}$ cup of sugar. If you are making $\frac{2}{4}$ of the recipe, how many cups of sugar do you need?

$$\frac{2}{2} \times \frac{2}{4}$$

→ Multiply the fractions: $\frac{2}{2} \times \frac{2}{4}$.

→ Numerator: $2 \times 2 = 4$. Denominator: $2 \times 4 = 8$.

→ Simplify: $1/2$ cups.

Answer: $\frac{4}{8} = 1/2$

12. Multiply the fractions and simplify: $\frac{2}{2} \times \frac{6}{5}$.

$$\frac{2}{2} \times \frac{6}{5}$$

→ Multiply numerators: $2 \times 6 = 12$.

→ Multiply denominators: $2 \times 5 = 10$.

→ Simplify $12/10$: $6/5$.

Answer: $\frac{12}{10} = 6/5$

13. Multiply: $\frac{3}{4} \times \frac{5}{4}$. Simplify your answer.

$$\frac{3}{4} \times \frac{5}{4}$$

→ Multiply numerators: $3 \times 5 = 15$.

→ Multiply denominators: $4 \times 4 = 16$.

→ Simplify $15/16$: $15/16$.

Answer: $\frac{15}{16} = 15/16$



14. A recipe calls for $\frac{2}{2}$ cup of sugar. If you are making $\frac{3}{2}$ of the recipe, how many cups of sugar do you need?

$$\frac{2}{2} \times \frac{3}{2}$$

→ Multiply the fractions: $\frac{2}{2} \times \frac{3}{2}$.

→ Numerator: $2 \times 3 = 6$. Denominator: $2 \times 2 = 4$.

→ Simplify: $\frac{3}{2}$ cups.

Answer: $\frac{6}{4} = \frac{3}{2}$



Percent of a number

15. Find 75% of 120.

75% of 120

→ Convert: $75\% = 75/100$.

→ Multiply: $75/100 \times 120 = 90$.

Answer: $75\% \times 120 \div 100 = 90$

16. A shirt costs \$120. It is 10% off. How many dollars is the discount?

10% of 120

→ Discount = 10% of \$120 = $10/100 \times 120 = \$12$.

Answer: $10\% \times 120 \div 100 = 12$

17. A store is offering a 41% discount on an item priced at \$88. How many dollars is the discount?

41% of 88

→ Discount = $41\% \times \$88$.

→ Convert percent: $41/100 = \text{decimal}$.

→ Discount = $41 \times 88 / 100 = \$36$.

Answer: $41\% \times 88 \div 100 = 36$

18. Find 75% of 60.

75% of 60

→ Convert: $75\% = 75/100$.

→ Multiply: $75/100 \times 60 = 45$.

Answer: $75\% \times 60 \div 100 = 45$

19. A shirt costs \$200. It is 50% off. How many dollars is the discount?

50% of 200

→ Discount = 50% of \$200 = $50/100 \times 200 = \$100$.

Answer: $50\% \times 200 \div 100 = 100$

20. A store is offering a 40% discount on an item priced at \$51. How many dollars is the discount?

40% of 51

→ Discount = $40\% \times \$51$.

→ Convert percent: $40/100 = \text{decimal}$.

→ Discount = $40 \times 51 / 100 = \$20$.

Answer: $40\% \times 51 \div 100 = 20$



Solving proportions

21. Solve the proportion: $2/4 = 2/x$. Find x .

$$\frac{2}{4} = \frac{2}{x}$$

→ Cross-multiply: $2x = 4 \times 2 = 8$.

→ Divide: $x = 8 / 2 = 4$.

Answer: $x = 8 \div 2 = 4$

22. A car travels 11 miles on 2 gallons of gas. How many miles can it travel on 6 gallons? Set up a proportion $2/11 = 6/x$ and solve.

$$\frac{2}{11} = \frac{6}{x}$$

→ $x = (11 \times 6) / 2 = 66 / 2 = 33$ miles.

Answer: $x = 66 \div 2 = 33$

23. If 6 shirts cost \$37, how much do 7 shirts cost? Set up and solve a proportion.

$$\frac{6}{37} = \frac{7}{x}$$

→ Set up the proportion: $6/37 = 7/x$.

→ Cross-multiply: $6x = 259$.

→ Divide both sides by 6: $x = 259/6 = 259/6$.

Answer: $x = 259 \div 6 = 259/6$

24. Solve the proportion: $4/6 = 3/x$. Find x .

$$\frac{4}{6} = \frac{3}{x}$$

→ Cross-multiply: $4x = 6 \times 3 = 18$.

→ Divide: $x = 18 / 4 = 9/2$.

Answer: $x = 18 \div 4 = 9/2$

25. A car travels 19 miles on 1 gallons of gas. How many miles can it travel on 5 gallons? Set up a proportion $1/19 = 5/x$ and solve.

$$\frac{1}{19} = \frac{5}{x}$$

→ $x = (19 \times 5) / 1 = 95 / 1 = 95$ miles.

Answer: $x = 95 \div 1 = 95$

26. If 2 shirts cost \$13, how much do 10 shirts cost? Set up and solve a proportion.

$$\frac{2}{13} = \frac{10}{x}$$

→ Set up the proportion: $2/13 = 10/x$.

→ Cross-multiply: $2x = 130$.

→ Divide both sides by 2: $x = 130/2 = 65$.

Answer: $x = 130 \div 2 = 65$



Writing and simplifying ratios

27. Write the ratio 12 to 36 in simplest form.

12 : 36

→ Write as a ratio: 12:36.

→ Find the GCF of 12 and 36: $GCF = 4$.

→ Divide both parts by 4: 3:9.

Answer: $\div 4 \Rightarrow 3 : 9$

28. A bag contains 2 red marbles and 6 blue marbles. Write the ratio of red to blue in simplest form.

2 : 6

→ Ratio of red to blue: 2:6.

→ GCF of 2 and 6 = 2.

→ Simplest form: 1:3.

Answer: $\div 2 \Rightarrow 1 : 3$

29. Write the ratio 24 to 9 in simplest form.

24 : 9

→ Write as a ratio: 24:9.

→ Find the GCF of 24 and 9: $GCF = 3$.

→ Divide both parts by 3: 8:3.

Answer: $\div 3 \Rightarrow 8 : 3$

30. A bag contains 12 red marbles and 4 blue marbles. Write the ratio of red to blue in simplest form.

12 : 4

→ Ratio of red to blue: 12:4.

→ GCF of 12 and 4 = 2.

→ Simplest form: 6:2.

Answer: $\div 2 \Rightarrow 6 : 2$

